What Single-Channel and Low-Cost EEG Can and Cannot Measure: A Reproducible Degradation Benchmark and Reporting Protocol

Eyyüb Güven*

Draft v4.0 — November 6, 2026

Abstract

Cheap EEG headsets now appear in classrooms, clinics and living rooms, often with claims about attention, relaxation and emotion. The hardware is capable; the reporting around it is not. A recording made with one dry electrode, a low sampling rate and an unknown noise floor cannot be interpreted without knowing how each of those choices changes the analysis. We built a benchmark that starts from open high-density recordings and walks them down, step by step, to a single-channel consumer setting: one electrode, 128 Hz, 1 µV quantisation, 1–40 Hz bandwidth, pink noise at 20/10/5/0 dB and synthetic blinks at 5/15/30 per minute. We then asked which standard analyses survive. In all 109 subjects of the EEG Motor Movement/Imagery Dataset, alpha reactivity was almost unchanged by the acquisition chain when the reference came from the same electrode (99.99% retained; 95% CI [0.9996, 1.0001]), while choosing a frontal site cost about 42% of eyes-closed alpha power (Fz/posterior ratio 0.58). Noise mattered: at 0 dB the effect fell to 59% of the reference. Our additive blink model changed the estimate by about 0.2%, but it cannot test real ocular artifacts. In 18 high-end recordings from an open affective dataset, valence, arousal and four-quadrant decoding were above chance with the full montage (0.69, 0.68, 0.58). With one electrode, valence was no longer detectable (0.52; equivalence untested), arousal weakened (0.56; $q = 0.050$ in the six-contrast family) and four-class accuracy fell to 0.34 (above the 0.25 chance level but far below the montage). None of four midline sites rescued valence, and a band control showed that part of the montage advantage depends on beta-band features, possibly muscle activity. A 14-channel consumer headset performed similarly to the matched high-end montage (0.67/0.71 versus 0.73/0.66 on the 12 shared participants), although a ±10-point equivalence test did not pass. Short-lag alpha stability was low ($r_{SB} = 0.016$); block-level alpha power was stable (ICC(2,1)= 0.92). The motor arm was invalid because its reference decoder did not exceed chance. We provide the code, derived results, decision matrix and reporting checklist.

Keywords: consumer EEG, single-channel, reproducibility, benchmark, affective decoding, reporting standards

1 Introduction

A device in search of a method

Low-cost EEG headsets are sold as attention trainers, relaxation aids, sleep gadgets and emotion sensors. Some of them record real neural signal: consumer devices have been validated against

*Cyber Cognition Lab (pre-incorporation), Çorlu, Türkiye. eyyub.gvn@gmail.com. ORCID 0009-0005-6017-8622.

research amplifiers for event-related potentials [18, 1], and resting-state comparisons show that spectral estimates and test–retest behaviour differ across systems [28, 5]. The problem is what happens after the recording. A paper may report that alpha power increased during relaxation without saying which electrode was used, how the signal was filtered, whether blinks were removed, or how the result compares with a conventional montage. A reader cannot tell whether the finding reflects a cognitive process or an artefact of the recording setup.

This matters for two reasons. First, the same claim can be produced by very different pipelines, so the result is difficult to reproduce. Second, a single frontal electrode picks up cortical activity together with eye movements and muscle noise [37]; presenting that mixture as a specific cognitive measure invites overclaiming. Device-comparison studies are useful, but they answer a different question: whether one headset is close enough to a reference amplifier under one protocol. They do not tell a reader which analyses still work when the acquisition is reduced.

What this paper does

We take high-quality open recordings and degrade them one step at a time towards what a cheap headset provides. The steps are explicit: a single electrode, 128 Hz sampling, 1 µV quantisation, 1–40 Hz band-limiting, pink noise at four levels, and synthetic blinks at three rates. Each step is reported separately, so a reader can see which manipulation matters. The benchmark has four analysis families: alpha reactivity, affective decoding, motor decoding (exploratory) and reliability. The reference is always the same recording before degradation, and, where it matters, the same electrode rather than a whole-head average.

Two open datasets make this feasible. The EEG Motor Movement/Imagery Dataset provides 109 subjects, 64 channels and eyes-open and eyes-closed baselines [34]. An open affective dataset contains 20 participants recorded simultaneously with a low-cost Emotiv EPOC+ and a high-end BrainVision system, with valence–arousal quadrant labels [32, 31]. The first dataset is used for the degradation grid; the second lets us compare a real low-cost device with a high-end montage and with the same high-end data reduced to one electrode.

What we find

The results are mixed, and the mix is the point. Alpha reactivity is robust to the acquisition chain: once the reference is taken at the same electrode, 99.99% of the effect remains. The choice of electrode, however, costs about 42% of eyes-closed alpha power, and noise reduces the effect monotonically. In the affective data, the full montage decodes valence, arousal and quadrant above chance, but a single electrode does not: valence is no longer detectable, arousal weakens, and four-class accuracy drops close to the level expected from a weak classifier. A band control suggests that part of the montage advantage comes from beta-band features, which may include muscle activity. A 14-channel Emotiv performs about as well as the high-end montage in this sample, but the equivalence test does not pass, so the comparison remains descriptive.

The contributions are: a reproducible degradation pipeline; a decision matrix that maps each result to a claim-level verdict; a same-site retention estimate for alpha; a decomposition of the affective result into representation, site and band effects; and a reporting checklist for single-channel studies, based on COBIDAS MEEG [24], EEG-BIDS [25] and the CRED-nf checklist [30]. The code, the derived results and the JSON files behind every number are archived with the paper.

The paper is not a clinical or wellness intervention study, and it does not validate a commercial product. It is a methods study: it measures what different analyses can support, and it shows where the limits lie.

2 Related Work

Consumer and low-cost EEG validation

Comparisons of medical-grade and consumer wireless systems report that consumer hardware can recover common event-related components under favourable conditions, while signal quality and artifact susceptibility differ substantially between devices and sessions [28, 18, 33]. One of the closest prior studies compared four systems (including MindWave and Muse) on five participants across two visits, focusing on Fp1 and explicitly comparing eyes-closed and eyes-open spectra and their test–retest reliability [28]. That is close to our alpha arm on real hardware, so the gap cannot be "nobody has looked at consumer spectral reliability"; the gap is that existing work does not quantify which analyses survive a *controlled* degradation chain, and does not connect that chain to claim-level verdicts. Work on channel subsets for emotion decoding has shown that small, well-chosen montages can be competitive with large ones [38]; what is missing is a controlled degradation of the same recordings with a site-matched reference and a claim-level decision rule. Our contribution is therefore the decomposition and the decision matrix, not the observation that fewer channels can hurt.

Reproducibility and reporting standards

Psychology and neuroscience face a well-documented reproducibility problem [23]. In electrophysiology the community response has been standardised reporting (COBIDAS MEEG) [24] and standardised data organisation (EEG-BIDS) [25]. Both are written primarily for research-grade acquisitions; they specify what to report, but not what can be inferred when the acquisition is a single channel. The present benchmark is designed to fill that inference gap and to make the reporting checklist operational for consumer-device studies.

Analysis pipelines and artifact handling

Modern open pipelines make preprocessing explicit and reproducible [10, 8, 22], and automated artifact rejection and independent-component classification reduce researcher degrees of freedom [14, 27]. These tools assume enough channels for spatial decomposition; with one channel, ICA is unavailable and most artifact rejection reduces to amplitude thresholds or regression against auxiliary signals. A benchmark of single-channel conditions therefore must treat artifact handling as part of the independent variable, not as a preprocessing detail.

Feature validity and open data

Band-power features are the workhorse of consumer-EEG claims, and their test–retest reliability is a prerequisite for any individual-level interpretation. Open datasets with documented conditions and affective labels allow reliability and decoding to be measured rather than asserted [34, 17]; the labels are normative picture categories, not participants' own reports. We build directly on the EEGBCI resources and on an open affective dataset that contains both low-cost and high-end recordings [32, 31]; DEAP and DREAMER are discussed as alternatives that could not be used. We report every degradation step so that a reader can map our results onto their own acquisition.

The affective dataset used here has its own decoding study [32], which compared the low-cost and high-end systems under three channel-count and sampling-rate scenarios and reported F1 values for four-class classification. Our analysis is not a replication of that study: we use balanced accuracy, within-subject cross-validation, a fully crossed degradation chain that includes quantisation, noise and blink arms, a same-site reference, and claim-level decision rules. The F1 values in the source study and the balanced accuracies here are therefore not directly comparable, and we do not interpret our lower values as contradicting it. What this paper adds is the

decomposition of where performance goes when the representation is reduced, not a new headline accuracy for the dataset.

Positioning

Relative to device-comparison work, our contribution is analysis-centred and degradation-based. Relative to reporting standards, it is quantitative: rather than saying which variables to report, it estimates what each variable is worth under single-channel constraints. To our knowledge, no existing benchmark reports, for the same open datasets, how alpha reactivity, single-trial decoding and reliability change as a function of explicit consumer-hardware degradations, and which of those changes survive a claim-level decision rule. Blink susceptibility is a stated limitation of this version and is not claimed as a measured outcome.

Benchmarking methodology

The idea of comparing analysis pipelines on shared datasets is established in BCI research: MOABB provides trustworthy benchmarking across datasets and pipelines [15]. Our benchmark differs in object — it holds the pipeline approximately fixed and varies the acquisition conditions, rather than comparing algorithms — but it inherits the principle that a claim about a method should be evaluated against a common, open reference. Filter design is a concrete example of why this matters: apparently small high-pass choices can create artifactual effects and incorrect conclusions [36], which is exactly the kind of degradation our chain isolates and reports.

3 Methods

Datasets

EEG Motor Movement/Imagery Dataset [34]. All 109 subjects were used: 64 channels at 160 Hz, runs 1–2 for the eyes-open and eyes-closed baselines and runs 4, 8 and 12 for imagined left/right hand movement, as defined in the dataset documentation. Motor decoding uses the ten subjects for whom the imagery runs were degraded in this pass, and is treated as exploratory.

Affective dataset [32, 31]. The open CC BY 4.0 dataset from Zenodo 14787743 contains EEG from 20 participants who viewed International Affective Picture System images selected to induce four valence–arousal quadrants, recorded with a low-cost Emotiv EPOC+ and a high-end BrainVision system. The archive's `EEG/Experiment_files` directory contains 18 BrainVision recordings (66 channels, 500 Hz, sampling interval 2000 µs); a nineteenth recording sits in a separate `EEG registro_nuevo` folder and was not used. All 18 recordings in the analysed directory completed the pipeline. The files are named `eegbci<subject>-<session>` by the archive; this is the archive's own naming for the affective recordings and does not refer to the EEGBCI motor dataset. The valence/arousal labels come from the affective picture protocol, not from a motor task. BrainVision marker codes map to the quadrants as follows: 1 = high valence/high arousal, 2 = low valence/high arousal, 3 = high valence/low arousal, 4 = low valence/low arousal, 10/14 = neutral. Each recording contains approximately 10 trials per quadrant and 8 neutral trials; neutral trials are excluded from valence and arousal analyses. Quadrant labels are the normative assignments of the International Affective Picture System [20], not participants' own ratings. Marker positions are used as onsets and each trial is epoched 0–2.5 s after onset. The low-cost Emotiv arm is analysed for the 12 participants whose CSV files are usable in the distributed archive (five files are empty or header-only and three fail the timestamp alignment). The device has been used in validation work on event-related potentials [1]. The files contain 14 EEG channels at 128 Hz in device units with absolute wall-clock timestamps. The log session whose absolute trial times fall inside the CSV span is selected; trials are aligned by converting each log onset to an absolute time (SessionStartDateTimeUtc plus the millisecond onset) and matching it

to the nearest CSV sample, with a clock offset chosen from $\{0, \pm 1, \pm 2\}$ hours by maximising the number of trials inside the recording span. Emotiv epochs run from -0.2 to $+2.3$ s around the aligned sample (the BrainVision epochs run 0–2.5 s), and the same band-power and LDA pipeline is applied. The median absolute difference between a log onset and its nearest CSV sample was 25 ms (95th percentile 59 ms, maximum 70 ms; stored per subject in `results/emotiv.json`), and no trial exceeded the 1 s alignment screen (this is a timestamp screen, not an amplitude rejection threshold). The two condition arms in the archive are not otherwise distinguished. This yields a three-way comparison: high-end full montage, low-cost 14-channel headset, and high-end data reduced to a single degraded channel.

Both datasets are de-identified and openly available. No new data were collected.

Degradation grid

EEGBCI recordings are processed at a full-montage reference and at twelve single-channel conditions: a cumulative acquisition chain (Fz; +128 Hz; +1 µV quantisation; +1–40 Hz band-limiting), a noise arm (20/10/5/0 dB pink noise, no blinks; pink noise is a conservative broad-band proxy and real consumer noise also contains white electronics noise, mains interference and drift, so the noise arm should be read as one plausible contamination profile rather than a device model), a blink arm (5/15/30 per minute, no noise) and a combined arm (0 dB + 30 blinks/min). The single electrode is selected by canonical name matching because EEGBCI channel labels carry punctuation (e.g. `Fz.`); the chosen label is written to the condition metadata. Blink transients are 350 ms Hann-windowed 2 Hz templates scaled to four standard deviations of the channel; with one channel the template cannot be given a frontal topography, which is a limitation of the model.

For the affective dataset the same single-channel chain (128 Hz, 1 µV, 1–40 Hz, Fz) is applied to the high-end recording and compared with the full montage. The noise and blink arms are not applied to the affective data; that comparison is left to future work.

Site-matched sensitivity analysis

A single Fz electrode and a 64-channel average are not the same measurement: alpha is posterior-dominant, so comparing them confounds channel count with electrode site. We therefore compute, in the eyes-closed run, Fz power from the full-montage recording, Fz power after the full single-channel chain, the 64-channel average and the average over posterior channels. The site-matched ratio (single-channel Fz versus full-montage Fz) is the primary estimate of what the acquisition chain costs; the scalp-average comparison is reported as the naive estimate that a reader would get without site-matching.

Analyses

A. Alpha reactivity. Band power (8–12 Hz) is computed with Welch's method (2 s windows, 50% overlap) on continuous data; the outcome is the natural-log ratio of eyes-closed to eyes-open power, the classical resting-state contrast [2]. The primary retention estimator is the geometric mean ratio derived from the paired subject-level log difference. For the acquisition chain the reference is the same-site full recording (Fz in the full montage); for the noise, blink and combined arms the reference is the clean single-channel chain, because those manipulations are incremental contamination of that chain. A degradation claim is reported as *supported* when the *lower* 95% confidence bound of the relevant retention ratio exceeds 0.50; as *not supported* when the lower bound is at or below 0.50; and the decision matrix reports the retention interval alongside the verdict. The 50% threshold is a planned convention of this benchmark, not a literature-derived constant, and it is not a formal non-inferiority test. The pre-specified 60% practical floor for balanced accuracy is reported separately as a stricter auxiliary criterion. Absolute powers are released so readers can evaluate alternative summaries.

B. Motor decoding (exploratory). Single-trial log band power (8–30 Hz; mu/beta event-related desynchronisation [26]) with shrinkage LDA and 5-fold stratified within-subject cross-validation. A CSP-based reference decoder (CSP fitted inside each fold) is reported as a sensitivity analysis. Because CSP requires multiple channels, the single-channel condition cannot receive the same spatial filter; that asymmetry is part of the constraint quantified here. The arm is reported as invalid unless a reference decoder's 95% CI excludes chance (0.50). Because nominal chance levels can be exceeded by chance, interpretation uses empirical confidence intervals rather than the theoretical level [6, 4].

C. Affective decoding. For each recording, log band power in theta (4–8 Hz), alpha (8–12 Hz) and beta (13–30 Hz) is extracted per channel and trial; the classifier is shrinkage LDA with 5-fold stratified within-subject cross-validation. Three targets are decoded: valence (quadrants 1/3 versus 2/4), arousal (1/2 versus 3/4) and the four quadrants. The same pipeline is run (i) on the full montage, (ii) after each cumulative acquisition step (128 Hz, quantisation, band-limiting), (iii) on single sites Fz, Cz, Pz and Oz, and (iv) on the full montage with only theta+alpha features and with only beta features. The last two conditions separate a representation effect from a band/EMG effect. Chance is 0.50 for the binary targets and 0.25 for the four-class target; the empirical CI is the inferential criterion [6]. The low-cost Emotiv arm is analysed separately and compared with the high-end arm on the 12 participants present in both, using a TOST procedure for equivalence.

D. Reliability. Alpha power is computed in consecutive 2 s windows of the eyes-closed run; odd and even windows are correlated, and the Spearman–Brown formula is reported as a descriptive *short-lag stability* measure. Odd and even windows are about two seconds apart, so this is not a split-half reliability coefficient and is not directly comparable to the ICC. In the ten-subject subset with three repeated imagined-hand runs, ICC(2,1) [35] of alpha power across runs is reported as a block-level estimate. Neither is a test–retest coefficient across sessions, which is not identifiable in these datasets.

E. Blink susceptibility of band ratios. Pre-specified but not implemented; reported as untested.

Statistical analysis

Subject-level outcomes are summarised with means and 95% confidence intervals (t-distribution, $df = n - 1$). Condition effects are reported both as unpaired means and as paired contrasts against the reference for the same subjects. Benjamini–Hochberg FDR [3] is applied within the alpha condition family and within the affective family; both raw and adjusted values are released. Mixed-effects models are not used here, and the exploratory nature of the study is stated explicitly. Equivalence between the low-cost and high-end devices is tested with a two one-sided tests (TOST) procedure using a ± 0.10 balanced-accuracy margin [19].

Reproducibility

All degradation parameters, random seeds, dataset file hashes (written by `scripts/01_download.py`), analysis scripts and derived feature tables are included in the repository. Raw datasets are not redistributed; download scripts reproduce the exact files used. Figure source data are archived with the results JSON. Artifact handling is deliberately minimal: no Autoreject or ICA is applied, because single-channel data cannot support ICA and the benchmark is about the acquisition conditions rather than about cleaning; the ocular-artifact literature [11, 7] and the EMG-contamination literature [37, 9, 21] are discussed as limitations. Single-channel test–retest work reports eyes-closed relative-power ICCs of roughly 0.76–0.85 for a consumer device [29]; our short-lag estimate is a within-session stability measure and is not directly comparable.

4 Results

Channel reduction, bandwidth and quantisation do not degrade alpha reactivity

At the full montage the eyes-closed/eyes-open alpha log ratio was 0.91 (95% CI $[0.78, 1.04]$; $N = 109$). The Fz single-channel chain — single electrode, 128 Hz, 1 µV quantisation, 1–40 Hz band-limiting — gave 0.77 ($[0.65, 0.89]$). Comparing the single channel with the scalp average, the paired difference was -0.14 ($[-0.19, -0.09]$). The mean per-subject retention ratio was 0.90 ($[0.85, 0.96]$) and the geometric estimator (ratio of means) was 0.87; the two estimators differ because ratios are skewed and both are reported in `results/paired.json` and `inference.json`.

That comparison is confounded by electrode site. Two corrections are needed. First, the acquisition chain itself costs almost nothing to *reactivity*: keeping the same electrode (Fz in the full recording versus Fz after the chain), the paired difference was -0.0001 ($[-0.0004, 0.0001]$), i.e. **99.99%** of the reactivity effect was retained; the corresponding retention interval $[0.9996, 1.0001]$ is stored in `results/site_metrics.json` (`chain_retention`). Second, the electrode-site decision is not free: eyes-closed alpha power at Fz is only about **58%** of the posterior-channel average (ratio 0.58, $[0.52, 0.63]$). A single-channel device therefore pays for its site choice, not for its sampling or bandwidth. This also explains why 128 Hz sampling, 1 µV quantisation and a 40 Hz low-pass left the estimate essentially unchanged: the band versus quantisation contrast was -0.00012 (retention 0.9999, $p = 0.007$), which is negligible for 8–12 Hz Welch power and consistent with earlier filter work [36].

Table 1 reports the condition means and Figure 1 shows the degradation grid, while Figure 2 shows retention.

Table 1: Alpha reactivity by condition ($N = 109$; mean log ratio with 95% CI). Noise and blink arms are separate; the final row combines both.

Condition	Log ratio	95% CI
Full montage (64 ch average)	0.91	[0.78, 1.04]
Single channel (Fz) chain	0.77	[0.65, 0.89]
+ noise 20 dB	0.76	[0.64, 0.88]
+ noise 10 dB	0.70	[0.58, 0.83]
+ noise 5 dB	0.60	[0.48, 0.72]
+ noise 0 dB	0.38	[0.26, 0.50]
+ blinks 5/min	0.77	[0.65, 0.89]
+ blinks 15/min	0.77	[0.65, 0.89]
+ blinks 30/min	0.77	[0.65, 0.89]
+ noise 0 dB and blinks 30/min	0.39	[0.27, 0.51]

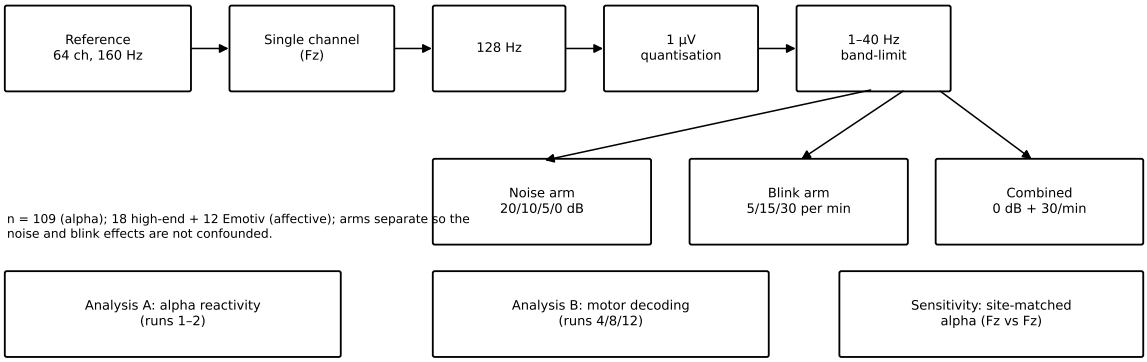


Figure 1: Degradation tree. The reference is the full-montage recording; the single-channel chain is cumulative, while the noise and blink arms branch from the fully degraded chain and are analysed separately.

Noise is the constraint that matters; blinks alone do not shift alpha reactivity

Noise produced a monotonic dose-response: 0.76 at 20 dB, 0.71 at 10 dB, 0.60 at 5 dB and 0.38 at 0 dB. Relative to the clean single-channel chain (the correct baseline for an incremental contamination), the geometric retention was 0.99, 0.94, 0.85 and 0.68; relative to the full-montage reference it was 0.86, 0.82, 0.73 and 0.59. The incremental paired contrasts against the clean chain (which isolate the noise manipulation) were -0.008 ($[-0.011, -0.006]$) at 20 dB, -0.064 ($[-0.075, -0.052]$) at 10 dB, -0.169 ($[-0.197, -0.140]$) at 5 dB and -0.384 ($[-0.441, -0.328]$) at 0 dB; all survive Benjamini–Hochberg correction across the condition family ($q < 0.001$). The three blink arms were almost identical to the clean chain (0.77 at 5 and 15/min, 0.77 at 30/min). The incremental contrast (30 blinks/min minus clean chain) was -0.002 ($[-0.003, -0.001]$; $q = 0.0005$), a change of about 0.2% that is statistically detectable but negligible in size; the combined arm (0.39) was driven by the 0 dB noise it contains. The injected template has almost no energy in 8–12 Hz and the same rate was added to both conditions, so this null is close to guaranteed by construction and should be read as a model check, not as evidence about real blinks; ocular artifacts are known to bias broadband power in ways this additive model does not represent [12]. The template has almost no 8–12 Hz energy and the same rate is added to both conditions, so the null is close to guaranteed and is not evidence about real ocular contamination [11, 7, 37].

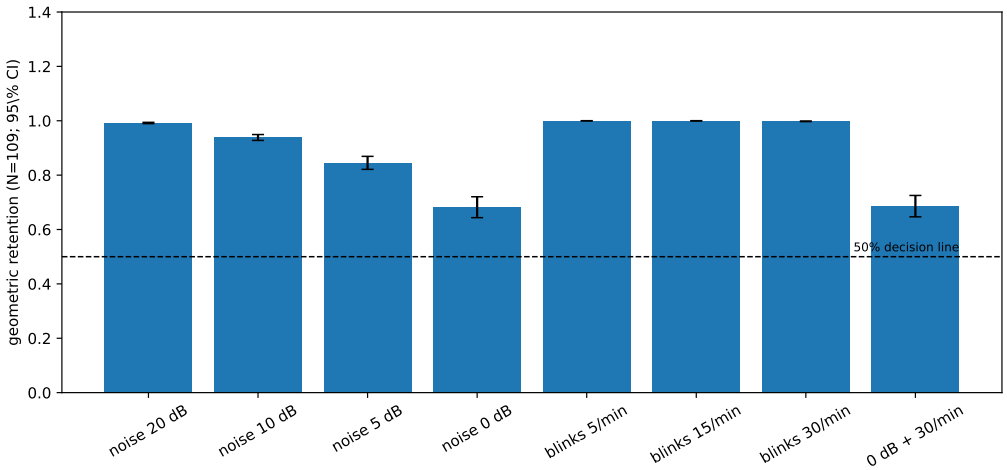


Figure 2: Alpha reactivity retained relative to the clean single-channel chain, by condition ($N = 109$; geometric retention and 95% CI). The dashed line is the planned 50% decision threshold. Noise reduces retention monotonically; the blink arm is a model check and is not a real-blink test.

Affective decoding survives a full montage but not one electrode

Table 2 reports the affective arm on the high-end BrainVision recordings (18 recordings from 20 participants; about 10 trials per quadrant). From the full montage, valence was decoded at 0.69 ($[0.62, 0.76]$), arousal at 0.68 ($[0.60, 0.75]$) and the four quadrants at 0.58 ($[0.49, 0.67]$); the binary intervals exclude chance and the four-class interval is above it. From the same data reduced to the single-channel chain, valence showed no significant evidence of above-chance decoding (0.52; $[0.47, 0.57]$, $p = 0.45$) — arousal to 0.56 ($[0.50, 0.62]$), which remains above chance but far below the montage, and the four quadrants to 0.34 ($[0.27, 0.41]$). The paired full-minus-single differences were $+0.17$ ($[0.10, 0.24]$) for valence, $+0.12$ ($[0.03, 0.20]$) for arousal and $+0.24$ ($[0.14, 0.34]$) for the four quadrants, so the collapse is not just a loss of significance.

A low-cost 14-channel Emotiv headset decoded the same targets at 0.67 ($[0.57, 0.77]$) for valence, 0.71 ($[0.63, 0.79]$) for arousal and 0.56 ($[0.46, 0.67]$) for the four quadrants. On the 12 participants present in both datasets, the low-cost minus high-end difference was -0.07 ($[-0.21, 0.08]$) for valence, $+0.05$ ($[-0.05, 0.15]$) for arousal and -0.04 ($[-0.21, 0.12]$) for the four quadrants. These differences are not statistically distinguishable from zero, but a ± 10-point equivalence test does not pass either (TOST $p = 0.31$, 0.14 and 0.23), so we describe the low-cost device as performing at a similar level in this sample, not that equivalence has been established.

Table 2: Affective decoding by acquisition representation: high-end full montage ($N = 18$ recordings), low-cost 14-channel Emotiv ($N = 12$), and high-end data reduced to one degraded channel ($N = 18$). Balanced accuracy with 95% CI; chance 0.50 (binary) and 0.25 (four-class).

Target	High-end montage	Low-cost 14 ch	Single channel
Valence	0.69 [0.62, 0.76]	0.67 [0.57, 0.77]	0.52 [0.47, 0.57]
Arousal	0.68 [0.60, 0.75]	0.71 [0.63, 0.79]	0.56 [0.50, 0.62]
Four quadrants	0.58 [0.49, 0.67]	0.56 [0.46, 0.67]	0.34 [0.27, 0.41]

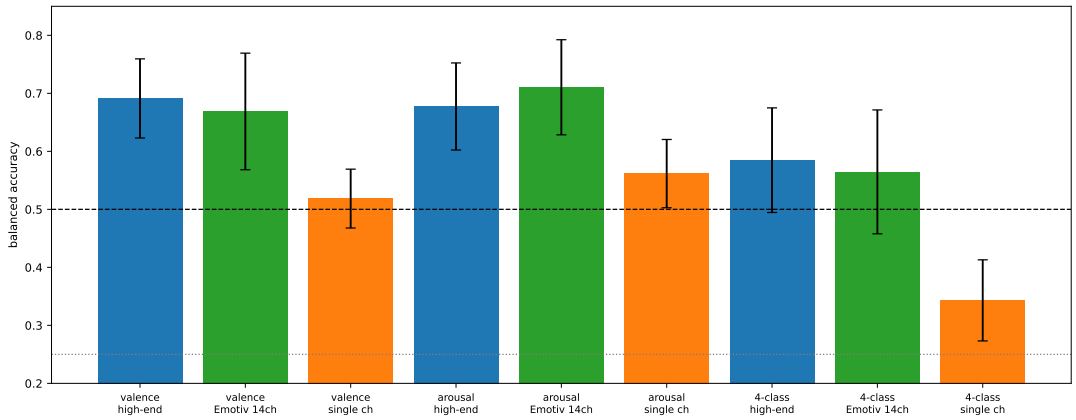


Figure 3: Affective decoding by acquisition representation: high-end montage, low-cost 14-channel Emotiv and single-channel chain. Dashed and dotted lines mark chance for binary (0.50) and four-class (0.25) targets.

Decomposing the affective collapse: representation, site and band

The chain that is harmless for alpha is also harmless for affective decoding (Table 3): resampling to 128 Hz, quantisation and band-limiting left four-class accuracy at 0.58–0.59 and valence at 0.69–0.70. The loss appears when the representation becomes one electrode. Valence was not distinguishable from chance at any of the four sites (0.51–0.53 at Fz, Cz, Pz and Oz); four-class accuracy was 0.33–0.38, above the 0.25 chance level but far below the montage; and arousal was weakly above chance at Fz, Pz and Oz before correction (0.56–0.57; unadjusted $p \approx 0.04$) but not at Cz (0.56, $[0.49, 0.62]$). The single-Fz arousal value survives the six-contrast affective family ($q = 0.0496$) but not the 12-test site sweep ($q \approx 0.079$); we therefore treat the site-level arousal pattern as exploratory. With only four midline sites and three features per site, the bounded claim is that none of these single-electrode representations approached the montage, not that every possible electrode would fail.

A band control changes the interpretation of the full-montage result. With only theta and alpha features the full montage decoded valence at 0.60 ($[0.54, 0.66]$) and the four quadrants at 0.46 ($[0.38, 0.53]$), whereas beta-only features retained 0.70 ($[0.61, 0.79]$) and 0.62 ($[0.53, 0.71]$). A substantial part of the full-montage affective advantage therefore sits in the beta band, where EMG contamination is a known concern [37, 9]. Because beta-only decoding was at least as high as the full montage (0.70 vs 0.69 for valence, 0.62 vs 0.58 for four-class), we do not interpret the montage advantage as purely additive cortical information, and we did not measure EMG directly. This is an interpretation, not a tested mechanism.

Table 3: Decomposition of affective decoding (18 high-end recordings; balanced accuracy). Chance is 0.50 for valence/arousal and 0.25 for four-class.

Representation	Valence	Arousal	Four-class
Full montage, all bands	0.69	0.68	0.58
Full montage, theta+alpha only	0.60	0.63	0.46
Full montage, beta only	0.70	0.70	0.62
128 Hz / quantised / band-limited	0.70	0.68	0.58
Single Fz	0.52	0.56	0.34
Single Cz	0.51	0.56	0.38
Single Pz	0.52	0.56	0.33
Single Oz	0.53	0.57	0.35

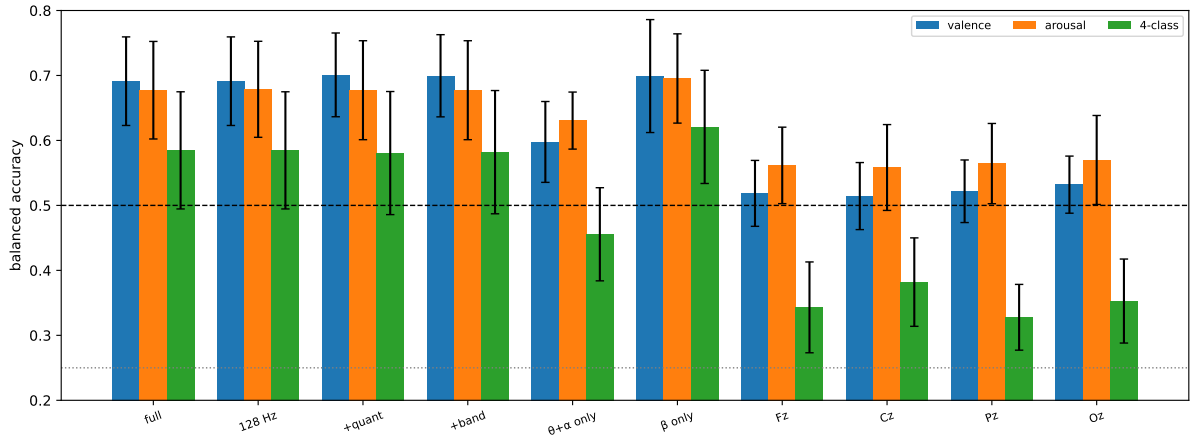


Figure 4: Decomposition of affective decoding by acquisition step and band. The dashed and dotted lines mark chance for binary (0.50) and four-class (0.25) targets.

Reliability depends on the timescale

Short-lag stability of consecutive 2 s alpha windows was $r_{SB} = 0.016$ ($[-0.08, 0.11]$; $N = 109$). This is a short-lag stability measure, not a classical reliability coefficient: odd and even 2 s windows are separated by about two seconds. It says that a momentary alpha estimate is noisy and is consistent with zero. In the ten-subject subset with three repeated imagined-hand runs, block-level alpha power had ICC(2,1)= 0.92, which captures stable between-subject differences. The two are not contradictory, and neither is a test–retest coefficient across sessions; published single-channel test–retest work reports ICCs of roughly 0.76–0.85 for eyes-closed relative power [29], but on a different design.

Exploratory motor decoding was inconclusive

In the ten-subject subset with imagery runs, balanced accuracy for imagined left/right hand was 0.51 ($[0.47, 0.56]$) at Fz and 0.54 ($[0.46, 0.62]$) at the full montage; a CSP reference reached 0.52 ($[0.41, 0.63]$) and a sensorimotor subset 0.55 ($[0.44, 0.66]$). Every interval includes chance, and the full-montage decoder has more features than trials (64 channels, 45 trials). Figure 5 shows the distribution of these accuracies. Because the reference decoder itself did not exceed chance, this arm cannot test the decoding hypothesis; it is reported as **invalid**, not merely inconclusive, and is not used for any claim.

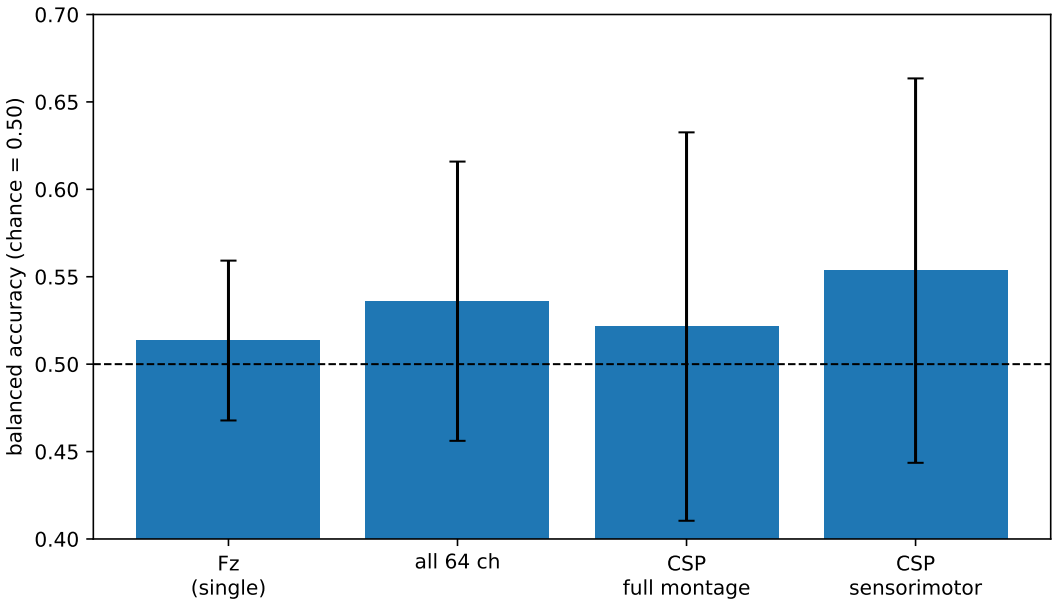


Figure 5: Exploratory motor-imagery decoding ($N = 10$), balanced accuracy by representation. All intervals include chance (0.50).

Decision matrix

Table 4 applies the planned 50% retention rule to the outcomes that were estimated.

Table 4: Claim-level verdicts.

Claim	Evidence	Verdict
Alpha reactivity survives the acquisition chain	Same-site reactivity retention lower bound 0.9996	Supported
Alpha reactivity survives 20–5 dB noise	Clean-chain retention lower bounds 0.99, 0.93, 0.82	Supported
Alpha reactivity survives 0 dB noise	Clean-chain retention 0.68 [0.64, 0.72]	Supported (attenuated)
Combined 0 dB + 30 blinks/min	Clean-chain retention 0.68 [0.65, 0.71]	Supported (attenuated)
Blinks bias alpha reactivity	Clean-chain retention 0.998 in a spectrally orthogonal, symmetric model	Not a real-blink test (H1 untested)
Pre-specified 60% balanced-accuracy floor	Valence full passes (lower 0.62); four-class full fails (lower 0.49); Emotiv valence fails (lower 0.57); arousal full/Emotiv pass	Mixed; reported as auxiliary criterion
Single-trial motor decoding above chance	Reference decoder did not exceed 0.50	Invalid (not testable)
Momentary individual-level alpha reliability	Short-lag stability $r_{SB} = 0.016$	Not supported
Stable individual alpha level	ICC across runs 0.92 (no CI)	Supported (block-level)

Status. The alpha arm uses the full 109-subject dataset. The affective arm uses 18 high-end recordings and 12 low-cost Emotiv recordings; the Emotiv alignment relies on wall-clock timestamps and 8 participants are dropped. The full-montage affective advantage is partly beta-band and may include a muscular contribution. The motor arm is invalid (reference at chance) and underpowered; H5 is untested. These scope limits are stated in the Discussion and are not treated as results.

5 Discussion

What survived, and what did not

The two halves of the benchmark give different answers. Alpha reactivity survives the acquisition chain almost intact: at the same electrode, the chain costs 0.01% of the effect. What costs alpha is the choice of electrode. Fz has about 58% of the eyes-closed alpha power of a posterior average, and this difference is a property of the scalp, not of the electronics. Noise also matters, and it matters in a predictable way: the effect declines steadily and reaches 59% of the reference at 0 dB. Bandwidth, sampling rate and quantisation, the specifications usually printed on a consumer headset's box, changed the 8–12 Hz estimate by a negligible amount.

Affective decoding behaves differently. With the full montage, valence, arousal and four-quadrant classification are above chance. With one electrode they are not: valence is undetectable, arousal is weak, and four-class accuracy falls to 0.34. The contrast is not a statement about broad alpha versus distributed affect. The two analyses use different effect sizes, different numbers of trials and different estimators, and the alpha ratio is partly invariant to the operations we applied. The useful conclusion is narrower: within this benchmark, the acquisition

chain does not limit alpha, but it does limit affective decoding once the spatial representation is reduced to one site.

Where the affective result goes

The decomposition helps. Resampling, quantisation and band-limiting leave full-montage accuracy unchanged. The loss appears when the data become one electrode, and it appears at all four midline sites we tried. Valence is not distinguishable from chance at Fz, Cz, Pz or Oz, and four-class accuracy stays between 0.33 and 0.38. Arousal is the exception: it remains weakly above chance at Fz, Pz and Oz before correction, though not after correction across the site sweep. We treat the site sweep as exploratory.

The band control changes how the montage result should be read. With only theta and alpha features, valence falls to 0.60 and four-class accuracy to 0.46. Beta-only features give 0.70 and 0.62, and beta-only is at least as good as the full montage. Beta is also where EMG contamination is most likely [37, 9]. We did not record EMG or test a paired beta-versus-theta/alpha difference, so the result is an interpretation: part of the montage advantage may be muscular rather than cortical. That possibility should be ruled out before the montage number is used as a neural benchmark.

The low-cost Emotiv arm is a useful sanity check. Its accuracy is close to the high-end montage on the 12 participants present in both datasets, and the matched differences are small. The equivalence test does not pass, however, so the appropriate description is "similar in this sample", not "equivalent". The comparison also mixes channel count, feature count, recording units, epoch window and condition arm, and eight participants are missing from the Emotiv analysis. It is a check, not a device validation.

What this means for reporting

The first lesson is about reference choice. Comparing one electrode with a whole-head average confounds channel count with electrode site, and a benchmark that does so will understate what a single channel retains. The correction is simple: compare the same site before and after degradation. Reference and site dependence of scalp power is not a new observation [13]; what is new here is the size of the effect for a single-channel benchmark, and the recommendation to make same-site comparison the default.

The second lesson is about the site itself. Fz is a convenient research midline electrode, but most single-electrode consumer devices record at the forehead: Fp1 in NeuroSky/MindWave and AF7/AF8 in Muse [18, 5]. Frontal sites have less posterior alpha and more ocular and muscle contamination, so the 42% site cost we measured is probably an underestimate for those devices. The consumer-EEG reliability comparison we cite is also at Fp1 [29], not Fz. The single-channel affective failure is established here for Fz, Cz, Pz and Oz, not for every possible electrode.

The third lesson is about timescale. A 2 s alpha window is a noisy estimate of a momentary state; the odd–even correlation is a short-lag stability measure, not a reliability coefficient, and its value near zero should not be read as "alpha is unreliable". Block-level alpha power was much more stable (ICC 0.92). A product that reports "your alpha right now" and a study that reports a stable individual difference are making different claims.

Limitations

The alpha arm uses all 109 subjects, but only the first 30 have the repeated motor runs needed for the ICC estimate; the ICC therefore rests on a small subset and has no confidence interval. The affective arm uses 18 high-end recordings, and the Emotiv arm covers 12 participants. The Emotiv signal is in device units, its epoch window differs slightly from the BrainVision epochs, and the archive's two condition arms are not distinguished. We did not measure a real single-channel

device, so the single-channel condition is a simulation of that device, not the device itself. The blink model is additive, spectrally orthogonal to alpha, and applied equally to both conditions; it cannot test the real ocular interaction, and the pre-specified blink hypothesis is therefore untested. Artifact handling is minimal: no ICA or Autoreject, so the beta-band features may be contaminated. The affective labels are normative IAPS categories rather than participants' own ratings, and we did not test picture-level generalisation or a label-permutation null. Only four midline sites were swept, and the single-channel condition has three features against 198 for the full montage, so the representation comparison confounds information with dimensionality. The 0 dB level is defined on broadband variance, the in-band signal-to-noise ratio is not reported, and the ratio estimator can compress the effect. Eight of 109 subjects had a negative reference reactivity, which is expected for a resting-state contrast but limits the precision of the mean. The analysis plan was not registered; several pre-specified items (Fp1/Cz sensitivity, the 60% floor, AUC and median-split outcomes) were changed or dropped, and the study is exploratory in the strict sense. The motor arm is invalid, not merely underpowered, because its reference decoder did not exceed chance. Finally, the equivalence test uses a small sample and a margin that is wide relative to the observed differences.

Where to go next

The obvious next step is a real single-channel device. A small study with a consumer headset, raw samples and the same pipeline would show whether the simulation is realistic, and whether the site cost is larger at the forehead, as we expect. A second step is to measure EMG or add a high-frequency control to separate muscular from cortical contributions to the beta-band result. A third is to register the analysis plan and repeat the alpha arm with a pre-specified decision rule. None of these requires new theory; they require better measurement, which is the point of the benchmark.

6 Conclusion

A single electrode does not automatically invalidate an EEG analysis. Alpha reactivity survived the acquisition chain almost completely when the comparison was made at the same site (99.99% retained), and the site itself, not the electronics, was the main cost. Affective decoding was less forgiving: it worked with a full montage and failed with one electrode, and part of the montage advantage appeared to come from beta-band features that may include muscle activity.

The practical output is a decision matrix, a reporting checklist and an analysis pipeline. The matrix is not a standard, and the study is exploratory: it uses simulated degradation, one real low-cost headset, and a limited set of sites. It does show that the question "can a cheap headset measure X?" has no single answer. It depends on the analysis, the electrode and the band. A benchmark that reports those dependencies is more useful than another device comparison, and easier to check.

Use of generative AI

Generative AI was used as an analysis and writing tool. The assistant (DeepSeek Harness, DeepSeek model) wrote and debugged analysis code, ran the statistical pipeline, generated figures and tables from the archived data, checked citation metadata against Crossref and DataCite, simulated reviewer feedback during revision, and edited the prose. Every number reported here was computed by the scripts in the repository and checked against the archived JSON; every cited reference was checked against a DOI record. The AI did not generate or alter data, did not produce any image presented as experimental data, and is not an author. The author takes full responsibility for the content and conclusions.

Data and code availability

All datasets are public. The EEG Motor Movement/Imagery Dataset is available from PhysioNet (doi:10.13026/C28G6P). The affective dataset is available from Zenodo under CC BY 4.0 (doi:10.5281/zenodo.14787743). DEAP was not used: its download page was unreachable during this work, and the DREAMER record is access-restricted [16]. The analysis repository (degradation pipeline, analysis scripts, seeds, environment file and the JSON files behind every number) is public at github.com/curiousbrutus/EEG-Benchmark and mirrored on the OSF project page osf.io/8wxqj. The archived version is deposited at Zenodo under DOI 10.5281/zenodo.23976432. The commit hash used for this draft is recorded in `results/data_manifest.json`. The single-channel reporting checklist is included as `CHECKLIST.md`. No new human participants were recorded.

Author contributions

Conceptualization, methodology, software, analysis, writing: E.G.

Acknowledgements

No funding was received for this work. The author declares no competing interests. Device-related collaboration and co-authorship will be declared with written permission if added.

References

[1] Nicholas A. Badcock, Kathryn A. Preece, Bianca de Wit, Kimberley Glenn, Nora Fieder, Johnson Thie, and Genevieve McArthur. Validation of the Emotiv EPOC EEG system for research quality auditory event-related potentials in children. *PeerJ*, 3:e907, 2015.

[2] Robert J. Barry, Adam R. Clarke, Stuart J. Johnstone, Christopher A. Magee, and Jacqueline A. Rushby. EEG differences between eyes-closed and eyes-open resting conditions. *Clinical Neurophysiology*, 118:2765–2773, 2007.

[3] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57:289–300, 1995.

[4] Martin Billinger, Ian Daly, Vera Kaiser, Jing Jin, Brendan Z. Allison, and Gernot R. Müller-Putz. Is it significant? guidelines for reporting BCI performance. In *Towards Practical Brain-Computer Interfaces*, pages 333–354. Springer, 2012.

[5] Céline Cannard, Helané Wahbeh, and Arnaud Delorme. Validating the wearable MUSE headset for EEG spectral analysis and frontal alpha asymmetry. In *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 3603–3610, 2021.

[6] Etienne Combrisson and Karim Jerbi. Exceeding chance level by chance: the caveat of theoretical chance levels in brain signal classification and statistical assessment of decoding accuracy. *Journal of Neuroscience Methods*, 250:126–136, 2015.

[7] Rodney J. Croft and Robert J. Barry. Removal of ocular artifact from the EEG: a review. *Neurophysiologie Clinique/Clinical Neurophysiology*, 30:5–19, 2000.

[8] Arnaud Delorme and Scott Makeig. EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, 134(1):9–21, 2004.

[9] Irina I. Goncharova, Dennis J. McFarland, Theresa M. Vaughan, and Jonathan R. Wolpaw. EMG contamination of EEG: spectral and topographical characteristics. *Clinical Neurophysiology*, 114:1580–1593, 2003.

[10] Alexandre Gramfort, Martin Luessi, Eric Larson, Denis A. Engemann, Daniel Strohmeier, Christian Brodbeck, Roman Goj, Mainak Jas, Teon Brooks, Lauri Parkkonen, and Matti Hämäläinen. MEG and EEG data analysis with MNE-python. *Frontiers in Neuroscience*, 7:267, 2013.

[11] Gabriele Gratton, Michael G. H. Coles, and Emanuel Donchin. A new method for off-line removal of ocular artifact. *Electroencephalography and Clinical Neurophysiology*, 55:468–484, 1983.

[12] Dirk Hagemann and Ewald Naumann. The effects of ocular artifacts on (lateralized) broadband power in the EEG. *Clinical Neurophysiology*, 112:215–231, 2001.

[13] Dirk Hagemann, Ewald Naumann, and Julian F. Thayer. The quest for the EEG reference revisited: a glance from brain asymmetry research. *Psychophysiology*, 38:847–857, 2001.

[14] Mainak Jas, Denis A. Engemann, Yousra Bekhti, Federico Raimondo, and Alexandre Gramfort. Autoreject: Automated artifact rejection for MEG and EEG data. *NeuroImage*, 159:417–429, 2017.

[15] Vinay Jayaram and Alexandre Barachant. MOABB: trustworthy algorithm benchmarking for BCIs. *Journal of Neural Engineering*, 15:066011, 2018.

[16] Stamos Katsigiannis and Naeem Ramzan. DREAMER: A database for emotion recognition through EEG and ECG signals from wireless low-cost off-the-shelf devices. *IEEE Journal of Biomedical and Health Informatics*, 22:98–107, 2018.

[17] Sander Koelstra, Christian Mühl, Mohammad Soleymani, Jong-Seok Lee, Ashkan Yazdani, Touradj Ebrahimi, Thierry Pun, Anton Nijholt, and Ioannis Patras. DEAP: A database for emotion analysis using physiological signals. *IEEE Transactions on Affective Computing*, 3(1):18–31, 2012.

[18] Olave E. Krigolson, Chad C. Williams, Angela Norton, Cameron D. Hassall, and Francisco L. Colino. Choosing MUSE: Validation of a low-cost, portable EEG system for ERP research. *Frontiers in Neuroscience*, 11:109, 2017.

[19] Dani"el Lakens, Anne M. Scheel, and Peder M. Isager. Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, 1:259–269, 2018.

[20] Peter J. Lang, Margaret M. Bradley, and Bruce N. Cuthbert. International affective picture system (IAPS): Affective ratings of pictures and instruction manual. Technical Report Technical Report A-8, University of Florida, 2008.

[21] Suresh D. Muthukumaraswamy. High-frequency brain activity and muscle artifacts in MEG/EEG: a review and recommendations. *Frontiers in Human Neuroscience*, 7:138, 2013.

[22] Robert Oostenveld, Pascal Fries, Eric Maris, and Jan-Mathijs Schoffelen. FieldTrip: Open source software for advanced analysis of MEG, EEG, and invasive electrophysiological data. *Computational Intelligence and Neuroscience*, 2011:156869, 2011.

[23] Open Science Collaboration. Estimating the reproducibility of psychological science. *Science*, 349:aac4716, 2015.

[24] Cyril Pernet, Marta I. Garrido, Alexandre Gramfort, Natasha Maurits, Christoph M. Michel, Elizabeth Pang, Riitta Salmelin, Jan Mathijs Schoffelen, Pedro A. Valdes-Sosa, and Aina Puce. Issues and recommendations from the OHBM COBIDAS MEEG committee for reproducible EEG and MEG research. *Nature Neuroscience*, 23, 2020.

[25] Cyril R. Pernet, Stefan Appelhoff, Krzysztof J. Gorgolewski, Guillaume Flandin, Christophe Phillips, Arnaud Delorme, and Robert Oostenveld. EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. *Scientific Data*, 6:103, 2019.

[26] G. Pfurtscheller and F. H. Lopes da Silva. Event-related EEG/MEG synchronization and desynchronization: basic principles. *Clinical Neurophysiology*, 110:1842–1857, 1999.

[27] Luca Pion-Tonachini, Ken Kreutz-Delgado, and Scott Makeig. ICLabel: An automated electroencephalographic independent component classifier, dataset, and website. *NeuroImage*, 198:181–197, 2019.

[28] Elena Ratti, Shani Waninger, Chris Berka, Giulio Ruffini, and Ajay Verma. Comparison of medical and consumer wireless EEG systems for use in clinical trials. *Frontiers in Human Neuroscience*, 11:398, 2017.

[29] Jeffrey M. Rogers, Stuart J. Johnstone, Anna Aminov, Joseph P. Donnelly, and Peter H. Wilson. Test-retest reliability of a single-channel, wireless EEG system. *International Journal of Psychophysiology*, 106:87–96, 2016.

[30] Tomas Ros, Stefanie Enriquez-Geppert, Vadim Zotev, Kymberly D. Young, Guilherme Wood, Susan Whitfield-Gabrieli, Feng Wan, Patrik Vuilleumier, Francois Vialatte, Dimitri Van De Ville, et al. Consensus on the reporting and experimental design of clinical and cognitive-behavioural neurofeedback studies (CRED-nf checklist). *Brain*, 143:1674–1685, 2020.

[31] Roberto Sánchez-Reolid. EEG dataset for emotion classification using low-cost and high-end equipment. https://zenodo.org/records/14787743, 2025. CC BY 4.0.

[32] Roberto Sánchez-Reolid, María Cruz Martínez-Sáez, Beatriz García-Martínez, Luz Fernández-Aguilar, Laura Ros, José Miguel Latorre, and Antonio Fernández-Caballero. Emotion classification from EEG with a low-cost BCI versus a high-end equipment. *International Journal of Neural Systems*, 32(10):2250041, 2022.

[33] Phattarapong Sawangjai, Supanida Hompoonsup, Pitshaporn Leelaarporn, Sirawaj Kongwudhikunakorn, and Theerawit Wilaiprasitporn. Consumer grade EEG measuring sensors as research tools: A review. *IEEE Sensors Journal*, 20:3996–4024, 2020.

[34] Gerwin Schalk, Dennis J. McFarland, Thilo Hinterberger, Niels Birbaumer, and Jonathan R. Wolpaw. EEG motor movement/imagery dataset. https://physionet.org/content/eegmmidb/, 2009.

[35] Patrick E. Shrout and Joseph L. Fleiss. Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86:420–428, 1979.

[36] Darren Tanner, Kara Morgan-Short, and Steven J. Luck. How inappropriate high-pass filters can produce artifactual effects and incorrect conclusions in ERP studies of language and cognition. *Psychophysiology*, 52:997–1009, 2015.

[37] Emma M. Whitham, Kenneth J. Pope, Sean P. Fitzgibbon, Trent Lewis, C. Richard Clark, Sally Loveless, Marita Broberg, Andrew Wallace, Dylan DeLosAngeles, Patrick Lillie, Andrew Hardy, Robert Fronsko, Alyson Pulbrook, and John O. Willoughby. Scalp electrical

recording during paralysis: quantitative evidence that EEG frequencies above 20 Hz are contaminated by EMG. *Clinical Neurophysiology*, 118:1877–1888, 2007.

[38] Wei-Long Zheng and Bao-Liang Lu. Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks. *IEEE Transactions on Autonomous Mental Development*, 7:162–175, 2015.